Dual-Channel Capsule Generation Adversarial Network for Hyperspectral Image Classification

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***Abstract*— Deep learning-based methods have demonstrated significant breakthroughs in the application of hyperspectral image (HSI) classification. However, some challenging issues still exist, such as the overfitting problem caused by the limitation of training size with high-dimensional feature and the efficiency of spectral–spatial (SS) exploitation. Therefore, to efficiently model the relative position of samples within the generative adversarial network (GAN) setting, we proposed a dual-channel SS fusion capsule generative adversarial network (DcCapsGAN) for HSI classification. Dual channels (1-D-CapsGAN and 2-D-CapsGAN) are constructed by integrating the capsule network (CapsNet) with GAN for eliminating the mode collapse and gradient disappearance problem caused by traditional GAN. Meanwhile, octave convolution and multiscale convolution are integrated into the proposed model for further reducing the parameters of the CapsNet and extracting multiscale features. To further boost the classification performance, the SS channel fusion model is constructed to composite and switch the feature information of different channels, thereby facilitating the accuracy and robustness of the whole classification performance. Three commonly used HSI data sets are utilized to investigate the performance of the proposed DcCapsGAN model, and the performance of the experiment demonstrates that the proposed model can efficiently improve the classification accuracy and performance.**

***Index Terms*— Capsule networks (CapsNets), generative adversarial network (GAN), hyperspectral image (HSI), multiscale convolution.**

I. Introduction

HYPERSPECTRAL images (HSIs), gathered by hyperspectral imaging sensors, contain rich spectral information using hundreds of contiguous narrow spectral bands, which can provide a detailed spectrum regarding the physical properties of materials. Benefit from such rich spectral information, HSIs have been effectively utilized to discriminate the categories of different land covers and targets. Therefore, HSIs have demonstrated an important role in agriculture [1], military [2], [3], environmental monitoring [4], [5], and so on. However, several challenges in the practice application of classification have been exposed, such as limitation of labeled training samples with the high dimensionality of spectral information, which can result in the well-known problem "curse of dimensionality" (the Hughes phenomenon) [6]. Besides, the exploration of rich spectral and spatial information is also an important challenge for the classification of HSIs.

Dimensional reduction (DR) techniques, such as principal components analysis (PCA) [7], linear discriminant analysis (LDA) [8], and band selection (BS) [9], have been successfully exploited for avoiding "curse of dimensionality" problem. Meanwhile, various pixel-based classifiers are applied to HSI classification depend on the discriminant information provided by the rich spectral feature, such as maximum likelihood [10], support vector machine (SVM) [11], artificial neural networks [12], sparse representation classifier (SRC) [13], collaborative representation classifier (CRC) [14], and multinomial logistic regression [15]. However, the above classification algorithms only adopted spectral information. To further improve the classification accuracy, the 3-D Gabor filters [16] and 3-D morphological profile (MP) [17] were employed for spectral–spatial (SS) feature extraction. In [18], MPs were used to extract spatial features of HSIs, and SVM was usually used to obtain the final classification map. In [19], a method combining sparse representation and Markov random field was proposed to explore spatial correlation and improve classification performance. Extended morphological profiles (EMP) were used to extract features of the most important principle components in [20]. Differential morphological profiles (DMP) are defined as the difference of MP between successive scales [21]. Especially, the segmentation map was obtained by associating each pixel to the level where the DMP value of the corresponding pixel reaches the maximum value. Fang *et al.* [22] proposed a novel local covariance matrix (CM) representation method to fully characterize the correlation among different spectral bands and the spatial–contextual information of HSIs. In [23], an SS derivative-aided kernel joint sparse representation (KJSR-SSDK) for HSI classification was presented. Majority of these methods [16]–[23] explored window-based techniques to extract spatial features, which demonstrates excellent performance. Besides, an adaptive nonlocal SS kernel (ANSSK) was

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proposed in [24] to further exploit homogeneous SS features, which presents promising HSI classification results.

Of late, deep learning (DL)-based methods have demonstrated very promising performance for many image analysis tasks. Chen *et al.* [25] first introduced the concept of DL into hyperspectral data classification and exploit the stacked autoencoders (SAE) to get joint SS features. Similarly, the potential of deep belief networks (DBN) for HSI classification was investigated in [26]. In [27], the ability of the restricted Boltzmann machine (RBM) is designed to capture distribution among pixels at the hidden level for extracting features of each band in HSIs. Recently, convolutional neural networks (CNNs) and their extension have gained great attention, which uses a local connection to extract spatial information and reduces the number of parameters by weight sharing. For example, a deep CNN architecture is employed directly in the spectral domain with one convolution layer, one pooling layer, and two fully connected layers for HSI classification [28]. In [29], a novel deep framework based on pixel-pair features (PPFs) learned by deep CNN was proposed as a deep spectral classifier. After that, Chen *et al.* [12] proposed a 3-D CNN-based feature extraction that employs several convolutional and pooling layers to extract SS features from HSIs. Zhang *et al.* [30] proposed diverse region-based CNN, which simultaneously utilizes spectral information, spatial information, and semantic context-aware information in each pixel by using the joint representation of the diverse region. Chen *et al.* [31] combined the Gabor filters with convolutional filters to effectively extract spatial information, including edges and textures. In addition, [32] and [33] applied some extensions to CNN, such as deep residual networks and densely connected networks for HSI classification. In [34], an end-to-end SS residual network (SSRN) was designed to learn robust SS representations from HSI. Furthermore, for improving the representational ability for HSI, a CNN with multiscale convolution (MS-CNNs) and diverse metrics framework is developed in [35].

Although the DL-based method has made great progress in HSI classification [25]–[35], the DL-based model usually faces the problem of overfitting with limitation of labeled samples because the method based on DL needs a large number of training samples to train the model, while hand-operated labeling of pixels is quite time-consuming in the field of remote sensing. Therefore, recently, generative adversarial networks (GANs) usually include a generation model G and a discrimination model D to address the issue of overfitting by generating new samples and adversarial training [36]. A family of GAN-based models has been proposed for visual tasks, such as auxiliary classifier GAN (ACGAN) [37], deep convolutional GAN (DCGAN) [38], and categorical GAN (CatGAN) [39]. Besides, triple GAN (Triple-GAN) composed of a generator, discriminator, and classifier was proposed in [40]. In the field of HSI classification, Zhu *et al.* [41] first explored the effectiveness of GAN for HSI classification, which demonstrates superior performance than traditional CNNs in the case of limited training samples. In addition, generative adversarial networks and conditional random fields (GAN-CRF) are proposed to integrate semi-supervised DL with GAN [42]. CNN has shown excellent performance in GAN-based models [36]–[42], but it cannot accurately model the relative position between features, especially when the input data rotates. Besides, the max-pooling procedure adopted in CNN-based models is liable to lose many detailed features for visual tasks. Therefore, capsule networks (CapsNets) as a powerful alternative to CNNs are proposed by Sabour *et al.* [43]. Different from CNN, CapsNet encodes data relationship into a vector (not scalar) and adopts vector neuron instead of the scalar neuron, where the norm value and orientation of vector represent the probability of feature existence and the instantiation parameter (relative position of samples, rotation angle, and so on) of feature, respectively. Similar to MS-CNNs [35], Xiang *et al.* [44] developed a novel variation of CapsNet to encode the hierarchical features, named Multiscale Capsule Network (MS-CapsNet). Inspired by GAN and CapsNet, Jaiswal *et al.* [45] incorporate capsules within the GAN framework (CapsGAN) and provide guidelines for designing CapsNet discriminators. In [46], the concept of capsule units was redefined to become SS units specialized in HSI classification. Zhu *et al.* [47] introduced a modified CapsNet named Conv-Capsule and provide competitive results compared with SVM, CNN, and recurrent neural network (RNN). For HSI classification, 1-D-GAN composed of CNN is proposed in [41]. Caps-TripleGAN is proposed by exploring the 1-D TripleGAN [40] for sample generation with integrating CapsNet in a serial model for HSI classification [48].

In view of that single-channel methods are difficult to achieve fast convergence and high-quality texture restoration at the same time, several dual-channel-based methods were proposed to combine complementary information. For image super-resolution [49], a deep channel to extract the detailed texture information and a shallow channel to recover the overall outline of the original image are constructed. In the field of remote sensing, two channels of CNN were proposed to learn features from the spectral and spatial domains; the learned features are concatenated and fed to a fully connected layer for extracting joint SS features [50]. Yang *et al.* [51] extended DenseNet [52] to dual-channel modus; 1-D customized DenseNet and 2-D customized DenseNet are utilized to extract hierarchical spectral features and spatial correlation features in series, respectively. Hu *et al.* [53] designed a 2-D and 3-D dual-channel structures based on ConvLSTM [54] to explore the intrinsic spatial–spectral structure of HSI.

Therefore, inspired by the aforementioned works, in order to fully utilize rich SS information and efficiently exploit and generate more available training samples with contexture relation information, in this article, we propose a dual-channel SS capsule fusion GAN (DcCapsGAN) for HSI classification. In the proposed DcCapsGAN, 1-D-CapsGAN and 2-D-CapsGAN are proposed as spectral and SS classifiers by the dual-channel capsule fusion GAN model. Meanwhile, two generators (spectral generator and SS generator) are designed, respectively, for generating and extracting joint SS features, and two discriminators (spectral discriminator and SS discriminator), based on CapsNets, are exploited as discriminators in the proposed network. Compared with the

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previous work, the contributions and novelty of the proposed DcCapsGAN mainly lie in the following aspects.

1) We exploit an elegant way to cooperate CapsNet with GAN for exploring the SS relations with simultaneously generating and augmenting available pseudospectral and SS training samples.
2) A local to global fully multiscale spectral feature extraction and generation structure of 1-D-CapsGAN, in terms of cooperating the advantage of CapsNet in spectral relation exploitation and the advantage of GAN in pseudosample generation, is constructed for simultaneously exploiting the discriminant spectral feature and generating spectral pseudosamples.
3) Octave convolution and multiscale convolution are combined to the proposed 2-D-CapsGAN model for constructing low- and high-level SS feature interaction and reducing the redundancy of parameters in the capsule structure. The proposed 2-D-CapsGAN structure can effectively improve the classification efficiency and accuracy by adequately exploring pixel-contexture spatial features in terms of the multiscale and octave structure that is utilized in the proposed 2-D-CapsGAN framework.
4) Dual-channel (1-D-CapsGAN and 2-D-CapsGAN) fusion framework is designed to effectively exploit and fusion the SS information when considering spectral and spatial contexture relations, where CapsNet structure is constructed in the discriminator part of the GAN to maintain the SS details of the samples, as well as improving the stability of the proposed model.

In summary, the proposed structure 1-D-CapsGAN can provide a local–global spectral relation analysis and generation modal, and the 2-D-CapsGAN can exploit the pixel to contexture level feature as interacting in low- and high-frequency feature maps with simultaneously generating available SS pseudosamples. The spectral and SS features are efficiently integrated by the dual-channel fusion modal for improving the overall classification performance. We verified our proposed 1-D-CapsGAN, 2-D-CapsGAN, and DcCapsGAN frameworks on several well-known hyperspectral data sets revealed that the proposed framework exhibits promising results and performance than several relative state-of-the-arts algorithms even under the condition of limited training samples.

The rest of this article is organized as follows. Section II presents the background of this article. Section III is dedicated to the details of the proposed SS capsule fusion GAN framework. Section IV reported the three real HSIs used in the experiments, the experimental results, and the comparisons with other methods. Finally, the conclusions and discussions of this article are drawn in Section V.

II. Related Work

The main work of this article is based on CapsNet and GANs. In this section, we briefly review CapsNet and GAN.

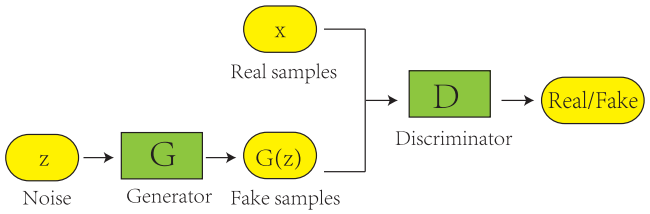


Fig. 1. Architecture of GAN.

A. Generative Adversarial Networks

GAN is a training method of the generative model, proposed by Goodfellow *et al.* [36]. As shown in Fig. 1, a typical GAN consists of two parts: a generator G that attempts to capture the distribution of real samples and generate the new samples closer to the distribution of the real sample by deconvolution. The discriminator is usually composed of traditional CNN to judge whether the input sample is real or generated by the generator. Because of the contradictory goal between generator and discriminator, the training process of GAN is to find and achieve the Nash equilibrium [55].

It is assumed that the real samples x have data distribution $p(x)$, while the input random Gaussian noise z has data distribution $p(z)$. The generator takes z as input and generates a mapping to data space $G(z)$. The input of the discriminator includes the real data x and the data $G(z)$ generated by the generator. The discriminator estimates the probability that the input derives from real data. The objective function of the GAN is given as follows:

$$\min_G \max_D V(D,G) = E_{x\sim p(x)}[\log(D(x))] + E_{z\sim p(z)}[\log(1-D(G(z)))] \quad (1)$$

where $V(\cdot)$ and E represent the observed value and the expected operator, respectively. $G(z)$ refers to the output of the generator. $D(x)$ and $D(G(z))$ refer to the output of the discriminator when the real samples x and generated samples $G(z)$ are input to the discriminator. During optimization, the discriminator and the generator try to maximize and minimize the function $V(D,G)$, respectively. The gradient of the discriminator should be fixed when training the generator; similarly, the gradient of the generator should be fixed when training the discriminator. Otherwise, the generator and the discriminator just need to simply cater to each other to reduce the value of the objective function rapidly, which makes the generator unable to learn the distribution of the real data well.

B. Capsule Networks

CapsNet is an improvement of traditional CNNs, which was proposed by Sabour *et al.* [43]. The core idea of CapsNet is that the output of neurons is not a scalar representation of activation but the vector that identifies the existence of feature entities and encodes their attributes into vector output. Compared with the traditional CNN, CapsNet has better feature extraction ability and generalization ability, which is suitable for tasks with a limited number of samples.

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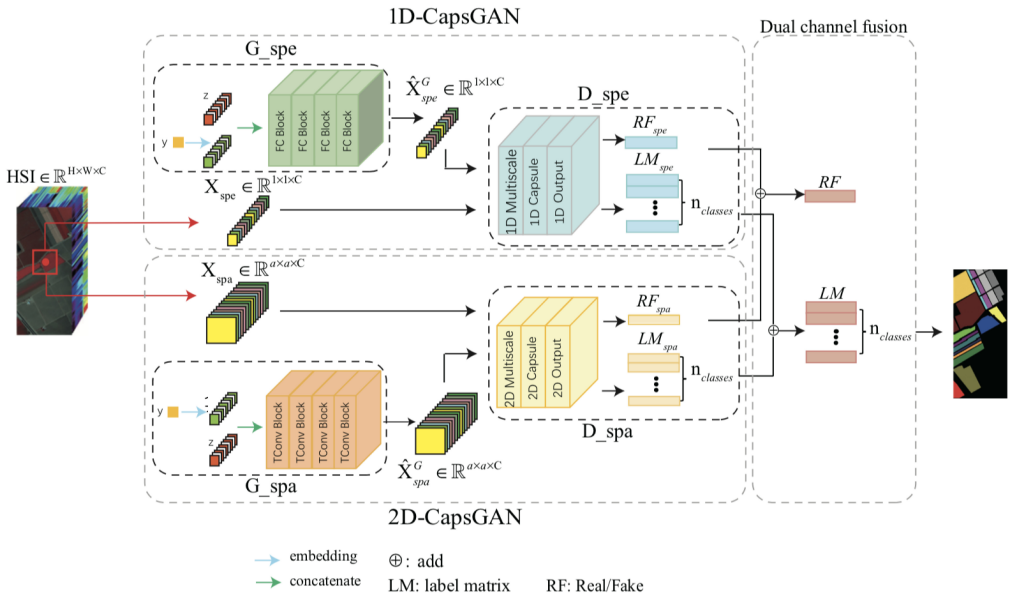


Fig. 2. Architecture of DcCapsGAN.

CapsNet also includes a routing-by-agreement mechanism for information transmission between capsules in continuous layers. For each capsule in layer L and each capsule in layer $L+1$, the coupling coefficient is adjusted iteratively according to the vector consistency (cosine similarity) between the current output of each capsule in layer L and the prediction of layer $L+1$. For classification tasks involving k classes, the last layer of CapsNet contains k capsules, each representing a class. The length of the capsule vector output represents the probability of the existence of the visual entity.

III. Proposed Method

A. Overview of the Proposed DcCapsGAN

CapsNet has been successfully applied to multiple views and application tasks because of its vectorization feature representation ability and can encode the rotation angle, relative position, and other instantiation parameters of features. Some existing methods [43]–[48] have proved that CapsNet can extract more relevant and complete information than traditional CNN. Different from conventional CNN, CapsNet utilizes vectors instead of scalars to represent the intensity of features, which efficiently enriches the representation and exploitation of existing features. For the classification of HSI, the discriminant information of spectral feature usually plays the key role for HSI classification; however, it is liable to cause a quantity of random discrete misclassified pixels in classification maps by only using spectral information. Spatial contexture information is usually involved to acquire smoother classification results for promoting classification performance and accuracies. Therefore, a dual-channel model of spectral and SS is designed to retain both high-level spectral information and spatial information. In addition, the proposed dual-channel capsule GAN structure can integrate the spectral and spatial features with simultaneously generating spectral and SS pseudosamples for alleviating the overfitting problem caused by the limitation of labeled samples in HSI classification.

The whole framework of the proposed DcCapsGAN method is shown in Fig. 2, which mainly contains the following parts: the 1-D-CapsGAN structure, which is composed of 1-D spectral pseudosamples generator and 1-D multiscale spectral discriminator; the 2-D-CapsGAN structure, which contains 2-D SS pseudosamples generator and 2-D SS discriminator, where octave convolution and multiscale convolution are adopted in discriminator part for eliminating the computational redundancy of capsules and improving the exploitation of spatial contexture. In the final parts, the fusion structure is constructed by the added model to make a fusion decision of 1-D-CapsGAN and 2-D-CapsGAN for further improving the classification accuracies. The abovementioned parts will be illustrated in detail in the following.

B. 1-D-CapsGAN

For HSI classification, the abundant spectral information and the relation between adjacent spectral features are the key factors to achieve classification by spectral information, so spectral classification is an important research direction of HSI. 1-D-CapsGAN is constructed based on the HSI spectral features to fully model multiclass spectral features and only use the spectral vector of the training data to train the model.

1) Spectral Feature Generator G_spe: As shown in Fig. 3(b), in spectral generator G_spe, all the input noise and training data are spectral vectors, G_spe receives both noise z and labels y as input, and the labels y are transformed into vectors by embedding operation. The embedded vector and noise z are fused by point multiplication. Four FC blocks are stacked in G_spe; batch normalization (BN) is used in each layer except the last one to improve generator stability.

2) Spectral Feature Discriminator D_spe: Most CNN-based methods usually retain the most representative

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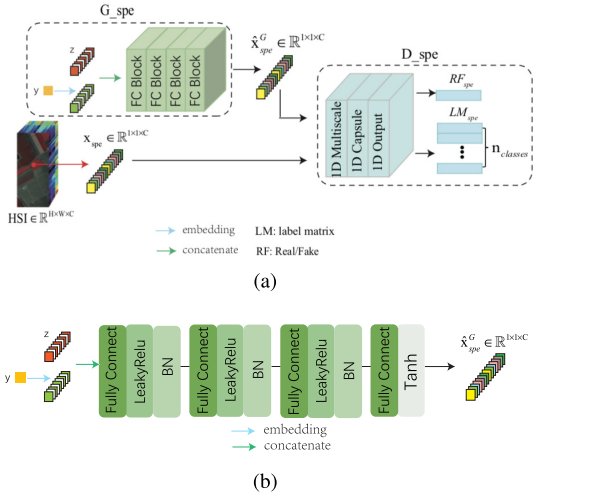


Fig. 3. 1-D-CapsGAN. (a) Overall structure of 1-D-CapsGAN. (b) Spectral generator.

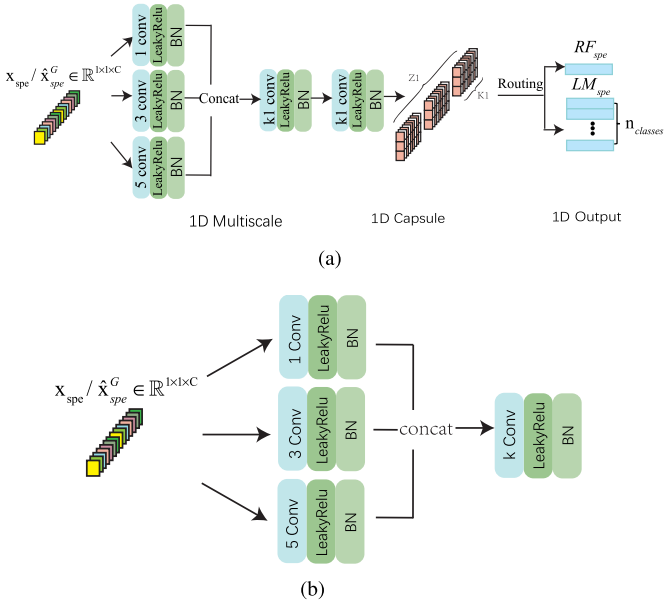


Fig. 4. 1-D Spectral multiscale discriminator in 1-D-CapsGAN. (a) Spectral discriminator. (b) Spectral multiscale structure.

frequency bands in HSI by dimension reduction operation and discard the frequency bands, which has little contribution to classification, which will inevitably lead to information loss. Therefore, in this article, CapsNet is explored instead of CNN in the discriminator part to learn a more robust representation of features. CapsNet encodes the attributes of feature into vectors, not scalars, which enables it to capture the relative position relation of spectral vectors and enhances the discriminant characteristics of bands.

As shown in Fig. 4(a), the proposed discriminator D_spe with 1-D spectral multiscale convolution is designed to exploit local–global spectral feature change and present more discriminate spectral feature. Spectral discriminator D_spe first receives the real spectral vectors x_{spe} and pseudospectral vectors $\hat{x}^G_{spe}$ as input in the 1-D Conv layer; then, the spectral multiscale convolution with the 1-D filter of different scales is constructed to extract information in different range. As shown in Fig. 4(b), 1-D convolution kernels with different scales of 1, 3, and 5 are used to extract detailed variations in spectral features. The next 1-D convolution with a filter length of $K1$ is applied to change the number of channels of the feature map and further extracted features. Thereby, the overall change of spectral features and the discriminant details change in a local spectral region can be efficiently exploited by the 1-D spectral multiscale convolution.

The capsule layers are established with $Z1$ numbers of capsules, where each capsule contains $K1$ capsule units, and the capsule units in the same capsule share their weights with the others. In this way, $Z1 \times K1$ capsule units are generated, each capsule unit is a d-dimensional vector denoted as u, and its orientation and length represent the instantiation parameters and the probability that the input data belong to this category. In order to properly represent the feature, the norm of the vector u is usually reduced by the nonlinear compression function represented by (2). The first part of (2) is to compress the norm between 0 and 1, and the second part is to keep the orientation of the vector unchanged so that the norm of u is compressed to 0 to 1 without changing their orientation

$$\tilde{u} = \frac{||u||^2}{1+||u||^2} \cdot \frac{u}{||u||}. \tag{2}$$

3) Dynamic Routing: After calculating the output of the capsule layer, dynamic routing is utilized to connect the 1-D capsule layer and 1-D output layer, which can design a learning process that is better than traditional pooling methods. First, u was multiplied to a transformation matrix W_{mn} to connect the mth capsule of the 1-D capsule layer with the nth capsule of the 1-D output layer

$$\hat{u}_{n|m} = W_{mn}\tilde{u}_m. \tag{3}$$

Then, the final input of the nth capsule of the 1-D output layer is obtained by calculating a weighted sum of $\hat{u}_{n|m}$

$$s_n = \sum_{m=1}^{Z_1 \times K_1} c_{m,n}\hat{u}_{n|m} \tag{4}$$

where $c_{m,n}$ denotes the coupling coefficient between the capsules of two layers, which measures the probability that the mth capsule of the 1-D capsule layer activates the nth capsule of the 1-D output layer. The final output vector v_n is obtained by squashed s_n through (2); therefore, v_n and s_n have the same direction

$$v_n = \frac{||s_n||^2}{1+||s_n||^2} \cdot \frac{s_n}{||s_n||}. \tag{5}$$

Especially, $c_{m,n}$ in (4) means coupling coefficient, and the sum of all coupling coefficients of the mth capsule of the 1-D capsule layer must be 1. It is obtained by the "routing softmax" expressed by the following equation:

$$c_{m,n} = \frac{\exp(b_{m,n})}{\sum_i^{n_classes}\exp(b_{m,i})} \text{ with } \sum_i^{n_classes} c_{m,i} = 1 \tag{6}$$

where $b_{m,n}$ denotes the log prior probability that the mth capsule of the 1-D capsule layer will activate the nth capsule

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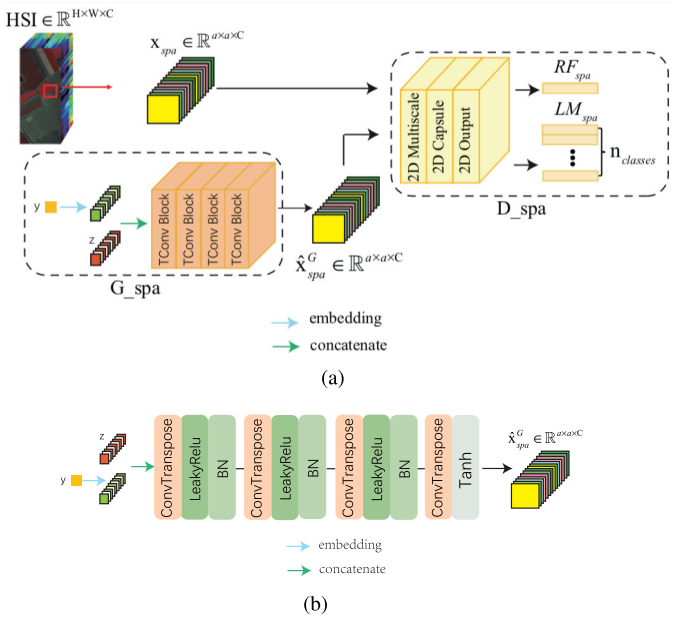


Fig. 5. 2-D-CapsGAN. (a) Overall structure of 2-D-CapsGAN. (b) SS generator.

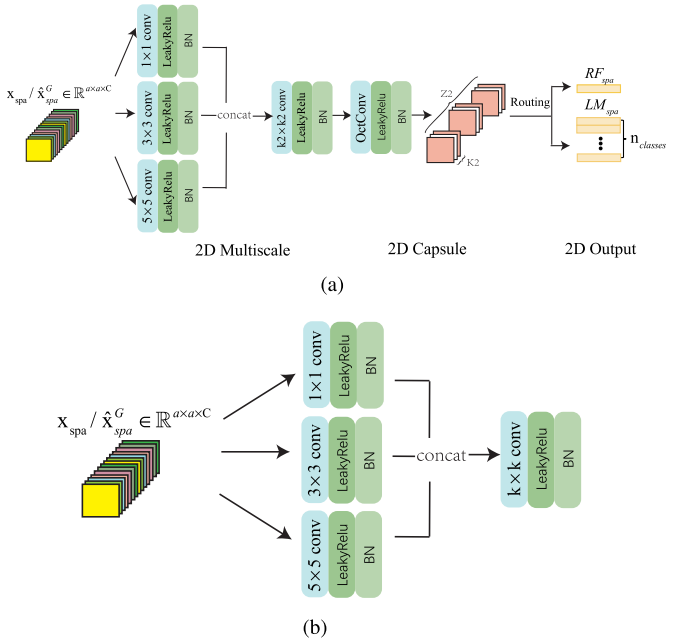


Fig. 6. 2-D Discriminator of 2-D-CapsGAN. (a) SS discriminator. (b) SS multiscale structure.

of the 1-D output layer. If the two capsules are very close in value, it can be concluded that they are highly correlated, and the correlation between the two capsules is measured by vector product. In this theory, $b_{m,n}$ is initialized to zero and can be updated in each iteration as follows:

$$b_{m,n} \leftarrow b_{m,n} + (v_n \cdot \hat{u}_{n|m}) = b_{m,n} + (|v_n||\hat{u}_{n|m}| \cos\theta). \quad (7)$$

As the iteration progresses, when v_n and $\hat{u}_{n|m}$ have high similarity, we can observe that $\cos(\theta) = \cos(0) = 1$, which increases $b_{m,n}$ and further increases $c_{m,n}$, s_n, and v_n.

The 1-D output layer aims at obtaining a collection of vectors v_n to effectively combine CapsNet and GAN and adjusts the model to HSI classification task; the proposed 1-D-CapsGAN outputs two parts: a vector RF_{spe} indicating the source of the input sample and a collection of n_{classes} vectors LM_{spe} where the ith vector is the capsule for class i. Since the norm of a capsule's vector output represents the presence of a visual entity, the norm of RF_{spe} indicates the probability whether the discriminator's input is a real or a generated vector, and the norm of ith vector in LM_{spe} is the probability of the input vector belongs to class i. At the training stage, the spectral feature generator G_spe and the spectral feature discriminator D_spe converge to a dynamic equilibrium, in which the generated data are gradually close to the real spectral data distribution.

C. 2-D-CapsGAN

Different from the 1-D-CapsGAN, we construct the 2-D-CapsGAN architecture to involve spatial information. The proposed architecture of 2-D-CapsGAN, the SS generator G_spa, and the SS discriminator D_spa structure are illustrated in Figs. 5(a), (b), and 6(a), respectively. The SS generator G_spa was designed to generate pseudo-SS samples by transposed convolutional (TConv) blocks [56], and the detailed architecture of TConv blocks is shown in Fig. 5(b). The structure of SS discriminator D_spa is designed according to 2-D-CapsNet, which is utilized to discern the samples real\fake and categories.

As shown in Fig. 5(b), it can be observed that a series of TConv are utilized, where the SS generator G_spa generates pseudo-SS samples with labeled information from the noise. After the training process of the generator and discriminator, the distribution of generated samples and training samples is more approximate to each other according to the proposed 2-D-CapsGAN architecture.

For D_spa, the 2-D-CapsNet is similar to 1-D-CapsNet that is explored mainly contains three parts, i.e., the 2-D conv layer, the 2-D capsule layer, and the 2-D output layer. In particular, to effectively extract and fuse multiscale image features, an SS multiscale structure using convolution kernels of different sizes is designed in a 2-D conv layer. Furthermore, octave convolution is integrated to reduce parameter redundancy in the proposed 2-D capsule layer. The same as the dynamic routing algorithm mentioned in part B, we utilized the dynamic routing algorithm in the last two capsule layers to calculate the output of capsule and make the network output a vector RF_{spa} to indicate the source of the input sample and a collection of n_{classes} vectors LM_{spa}, where the ith vector is the capsule for class i.

As shown in Fig. 6(b), in SS multiscale convolution, 2-D convolution kernels with spatial sizes of 1×1, 3×3, and 5×5 are employed to extract features of different scales with BN [57] and leaky rectified linear unit (LeakyRelu) activation functions. Then, the multiscale feature maps of different channels are concatenated to n times (n represents the numbers of scales) the number of channels. Subsequently, we reshape the concatenated feature maps of the same size

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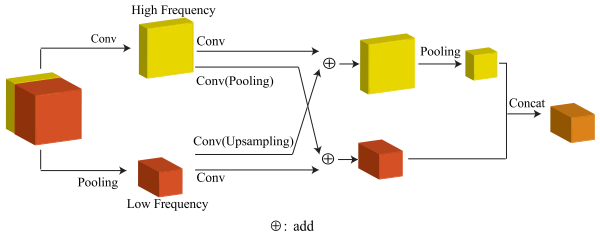


Fig. 7. Architecture of octave convolution.

by padding operation, where $k \times k$ convolution is applied to change the number of channels in the feature map and further extracted features. The customized SS multiscale convolution can enlarge the spatial receptive field of the model.

Octave convolution can reduce the spatial redundancy computation cost with better results in image classification through compressing low-frequency features while maintaining the high-frequency features [58]. The high-frequency component in feature extraction usually represents the details in one image (such as edge or object), and the low-frequency component usually depicts the overall shape of the image (such as slowly changed part). The architecture of octave convolution is shown in Fig. 7; we can observe that, through a kit of convolution, pooling, upsampling, and other operations in octave convolution, high-frequency information, and low-frequency information are efficiently merged and exchanged. We adopt octave convolution to eliminate the parameter redundancy problem of conventional CNN used in the capsule layer for further improving the prediction accuracies in 2-D-CapsGAN architecture.

D. Dual-Channel Fusion

Furthermore, for effectively fusing the output of 1-D-CapsGAN and 2-D-CapsGAN, the DcCapsGAN framework is designed, and RF_{spe} and RF_{spa} are added together to obtain RF, a joint prediction of the source of the input data, while LM_{spe} and LM_{spa} are added together to obtain the final label matrix LM, a joint prediction of the class to which the input data belong. Two-channel CapsGAN is utilized to generate pseudosamples and extracting spectral and SS information by combining the results of 1-D-CapsGAN and 2-D-CapsGAN.

The objective of the whole network architecture can be summarized as the following two parts: an adversarial loss as (8) and a margin loss for multiclass classification as (9):

$$\begin{aligned} L_{\text{adv}} &= \min_{G_\text{spe},G_\text{spa}} \max_{D_\text{spe},D_\text{spa}} V(D_\text{spe}, D_\text{spa}, G_\text{spe}, G_\text{spa}) \\ &= E_{x_{\text{spe}}\sim p(x_{\text{spe}}),x_{\text{spa}}\sim p(x_{\text{spa}})}\left[\log\left(\frac{D_\text{spe}(x_{\text{spe}}) + D_\text{spa}(x_{\text{spa}})}{2}\right)\right] \\ &\quad + E_{z\sim p(z)}\left[\log\left(1 - \frac{D_\text{spe}\left(\hat{x}_{\text{spe}}^{G}\right) + D_\text{spa}\left(\hat{x}_{\text{spa}}^{G}\right)}{2}\right)\right] \end{aligned} \tag{8}$$

$$\begin{aligned} L_{\text{margin}} &= \sum_{i}^{n_{\text{classes}}} \left(T_i \max\left(0, \alpha^{+} - ||v_i||\right)^2\right. \\ &\quad \left. + \lambda(1 - T_i)\max\left(0, ||v_i|| - \alpha^{-}\right)^2\right) \end{aligned} \tag{9}$$

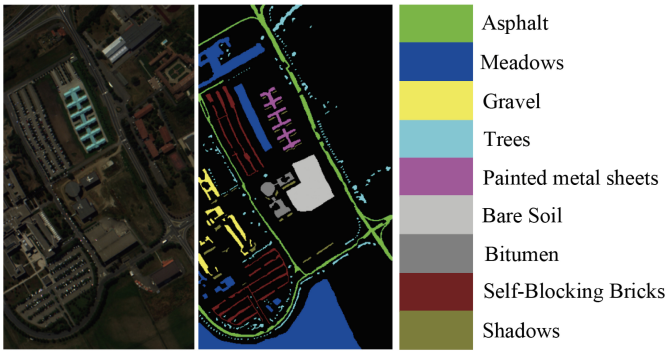


Fig. 8. Pseudocolor image and ground-truth map for the UP data set.

where v_i is the ith vector in Pre_{label}, and T_i is set to 1 if class i is present in the data and 0 otherwise. In addition, the parameters α^{+} and α^{-} as the boundary force the norm of the vector v_i to be in a smaller value range to avoid loss maximization or collapse. Suppose that $\alpha^{+} = 0.9$ and $\alpha^{-} = 0.1$; then, if the related classes exist, these boundaries force the length of vector v_i to be in the range of [0.9, 1], otherwise in the range of [0, 0.1]. In addition, λ is a regularization parameter, usually taken as 0.5, which reduces the influence of the vector that does not exist in the corresponding class.

The whole procedure of the proposed DcCapsGAN is summarized in Algorithm 1.

IV. Experiments

In order to verify the effectiveness, three real hyperspectral data sets are utilized to validate the performance of the method; we conducted experiments with ten randomly selected training data and evaluated the results by the average individual classification accuracy [%], the overall accuracy (OA) [%], the average accuracy (AA) [%], and the average and standard deviation of kappa coefficient (kappa) [%].

A. Hyperspectral Data Sets

1) University of Pavia: The University of Pavia (UP) data set is collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the UP in northern Italy. The data set contains 610×340 pixels. Each pixel has 103 spectral bands (after removing 12 very noisy bands, 103 spectral bands are left), the spectral band is between 430 and 860 nm, and the spatial resolution is 1.3 m/pixel. There are nine classes in the data set. Fig. 8 show the false-color image of the UP data set and the ground-truth map.

2) Salinas: The Salinas data set was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley. The data set is composed of 512×217 pixels with a high spatial resolution of 3.7 m/pixel. It contains 224 spectral bands; by removing 20 water absorption bands, the number of spectral bands is reduced to 204. The samples in the data set contain 16 different land cover classes. Fig. 9 illustrates the pseudocolor composite image and the ground truth of 16 classes.

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Algorithm 1 Procedure of the Proposed DcCapsGAN Method

Input: training samples $x = \{(x_{spe_i}, x_{spa_i})\}_{i=1}^{N}$ and test samples $x_{test} = \{(x_{spe_i}, x_{spa_i})\}_{i=1}^{S}$, the onehot labels $\{y_{onehot_i}\}_{i=1}^{N}$ of training samples, batch size n, the number of training epochs E, noise dimension d, the updating times k of the discriminators

Initialization: all the weight matrices and biases of *G_spe*, *G_spa*, *D_spe* and *D_spa*
for *E training iterations* **do**
  generating d dimensional noises $z = \{z_j\}_{j=1}^{d}$;
  input n spectral training samples $x = \{(x_{spe_i}, x_{spa_i})\}_{i=1}^{n}$;
  generating n spectral fake samples $\hat{\mathbf{X}}_{spe}^{G} = \{(\hat{x}_{spe_i}^{G})\}_{i=1}^{n}$ by spectral generator *G_spe* and n SS fake samples $\hat{\mathbf{X}}_{spa}^{G} = \{(\hat{x}_{spa_i}^{G})\}_{i=1}^{n}$ by SS generator *G_spa*;
  updating parameters of the G_{spe} and G_{spa} by minimizing $L_{adv} + L_{margin}$ through (8) and (9);
  for *k times* **do**
    extract multiscale spectral and SS features by 1-D multiscale and 2-D multiscale layers;
    convert the feature map into capsules in 1-D capsule layer and 2-D capsule layer;
    output RF_{spe}, LM_{spe}, RF_{spa}, LM_{spa} through the dynamic routing operation using formulas (2)–(7);
    updating parameters of the D_{spe} and D_{spa} by minimizing $L_{adv} + L_{margin}$ through (8) and (9);
  end
end
Output: the final label of the test samples x_{test} is depend on the largest norm of capsules in Pre_{label}

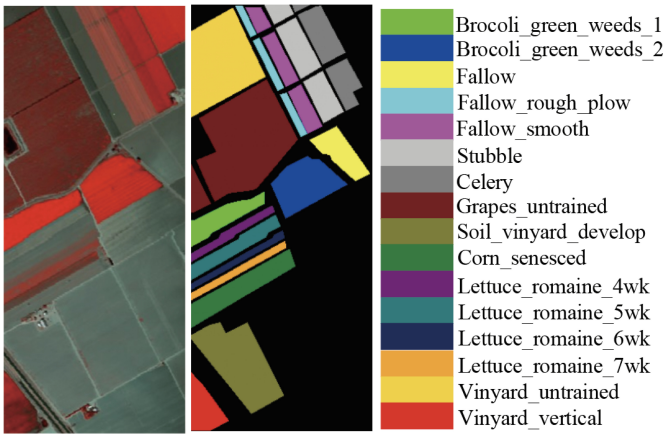


Fig. 9. Pseudocolor image and ground-truth map for the Salinas data set.

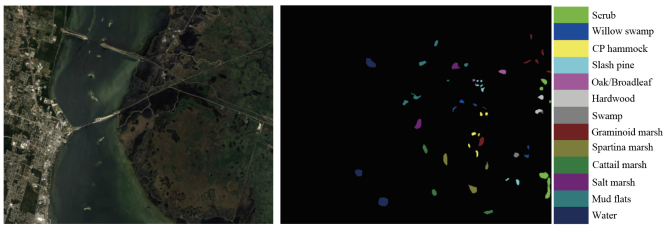


Fig. 10. Pseudocolor image and ground-truth map for the KSC data set.

3) Kennedy Space Center NASA: The Kennedy Space Center (KSC) data set was captured by the NASA AVIRIS instrument over the KSC. After removing the noisy channels, the KSC scene contains 176 spectral bands (from 400 to 2500 nm), 512 × 614 pixels, and the spatial resolution is 20 m/pixel, including 13 classes. Fig. 10 presents the pseudocolor composite image and the ground truth of the KSC data set.

B. Parameter Analysis

In this part, we mainly analyze the parameters' effect of the proposed DcCapsGAN for different data sets. In 1-D-CapsGAN and 2-D-CapsGAN, we mainly analyze the filter size in the 1-D/2-D multiscale layer, the number of capsules in the 1-D/2-D capsule layer, and vector dimensions of each capsule. In addition, the ablation experiments for the analysis of the effect of the octave and multiscale convolution in 2-D-CapsGAN are also presented in this part.

In our parameter analysis experiments, the batch size is set to 32. The Adam optimization algorithm is selected as optimizer [59], and the spatial size of 9 × 9 pixels for the input patches was considered. The number of capsules in the 1-D/2-D output layer is consistent with the number of categories, so, for Salinas, Pavia University, and KSC data sets, the number of capsules in the output layer is set to 16, 9, and 13, respectively. CapsNets have a large number of parameters because each capsule outputs a vector instead of a scalar and has additional parameters related to all the capsules in its upper layer; these additional parameters are used to predict the output of the upper layer.

First, the receptive field of the network is determined by the filter size, which represents the spatial scope that neurons can detect. As for the number of capsules in 1-D and 2-D capsule layers, we chose 8, 16, and 32 as the value scope; it notes that this value is the number of capsule groups, not the total number of capsules. CapsNets represent the scalars as vectors, so the length of each vector is particularly important. If the length of the vector is too short, it is liable to weaken the feature representation ability of the vector. If the length of the vector is too long, it is liable to decrease the effect of the model and the training efficiency. We choose 8, 16 as the value range to explore the effect of vector length on model performance in different data sets.

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TABLE I
NUMBERS OF TRAINING AND TESTING SAMPLING FOR THE UP DATA SET

Class		Samples		
No.	Name	Total	Training	Test
1	Asphalt	6631	40	6591
2	Meadows	18649	40	18609
3	Gravel	2099	40	2059
4	Trees	3064	40	3024
5	Painted metal sheets	1345	40	1305
6	Bare Soil	5029	40	4989
7	Bitumen	1330	40	1290
8	Self-Blocking Bricks	3682	40	3642
9	Shadows	947	40	907
-	Total	42776	360	42416

TABLE II
NUMBERS OF TRAINING AND TESTING SAMPLING FOR THE SALINAS DATA SET

Class		Samples		
No.	Name	Total	Training	Test
1	Brocoli_green_weeds_1	1977	20	1957
2	Brocoli_green_weeds_2	3726	20	3706
3	Fallow	1976	20	1956
4	Fallow_rough_plow	1394	20	1374
5	Fallow_smooth	2678	20	2658
6	Stubble	3959	20	3939
7	Celery	3579	20	3559
8	Grapes_untrained	11213	20	11193
9	Soil_vinyard_develop	6197	20	6177
10	Corn_senesced	3249	20	3229
11	Lettuce_romaine_4wk	1058	20	1038
12	Lettuce_romaine_5wk	1908	20	1888
13	Lettuce_romaine_6wk	909	20	889
14	Lettuce_romaine_7wk	1061	20	1041
15	Vinyard_untrained	7164	20	7144
16	Vinyard_vertical	1737	20	1717
-	Total	53785	320	53465

TABLE III
NUMBERS OF TRAINING AND TESTING SAMPLING FOR THE KSC DATA SET

Class		Samples		
No.	Name	Total	Training	Test
1	Scrub	761	25	736
2	Willow swamp	243	25	218
3	CP hammock	256	25	231
4	Slash pine	252	25	227
5	Oak/Broadleaf	161	25	136
6	Hardwood	229	25	204
7	Swamp	105	25	80
8	Graminoid marsh	431	25	406
9	Spartina marsh	520	25	495
10	Cattail marsh	404	25	379
11	Salt marsh	419	25	394
12	Mud flats	503	25	478
13	Water	927	25	902
-	Total	5211	325	4886

The effect results of different parameters are presented in Tables IV–VI, in which 1-D-CapsGAN and 2-D-CapsGAN are analyzed and explored, respectively. We can observe from Tables IV–VI the combination effect of different parameters on 1-D-CapsGAN and 2-D-CapsGAN; and for the UP data set, the optimal choice of the filter size, the number of capsules, and the vector dimension are 3, 8, and 8 for 1-D-CapsGAN and 3, 32, and 8 for 2-D-CapsGAN, respectively. For the Salinas data set, the optimal choice of the filter size, the number of capsules, and the vector dimension are 5, 16, and 8 for both 1-D and 2-D-CapsGAN. For the KSC data set, the best choice of the filter size, the number of capsules, and the vector dimension are 7, 8, and 16 for 1-D-CapsGAN and 5, 16, and 8 for 2-D-CapsGAN, respectively.

In order to verify the effects of multiscale convolution and octave convolution, as shown in Table VII, an ablation study is executed on the UP data set. We can observe that the convolutional layer in CapsNet with multiscale convolution to extract information of different scales can improve the performance of the model. At this time, the octave convolution can reduce the execution time and improve the classification accuracy.

C. Classification Results and Analysis

In this section, we compare our proposed method with the following relative competitive algorithms: 1) spectral-based SVM (SVM) [60]; 2) SVM with CK (SVMCK) [60]; 3) spatial-based 2-D-CNN (2-D-CNN) [12]; 4) SS-based 3-D-CNN (3-D-CNN) [61]; 5) spectral–spatial residual network (SSRN) [34]; 6) CapsNet [46]; and 7) GAN [41]. The numbers of training samples and test samples used in each data set are listed in Tables I–III. The classification results with mean and standard deviation values and the training time of different algorithms are summarized in Tables IV and V for the experimental data sets with ten times Monte Carlo random runs for reducing possible bias. The best results are labeled in Bold.

From Tables VIII–X, we can observe that SVMCK and DL-based methods, such as 2-D-CNN, 3-D-CNN, SSRN, CapsNet, GAN, and the proposed DcCapsGAN, are superior to SVM due to their hierarchical nonlinear feature extraction property. Compared with spectral or spatial features considered by SVM and 2-D-CNN, by joint involving SS features, the classification performance of SVMCK and 3-D-CNN has been greatly improved compared with SVM and 2-D-CNN;

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TABLE IV
CLASSIFICATION EFFECT OF D_SPE AND D_SPA UNDER DIFFERENT PARAMETERS FOR THE UP DATA SET

Filter size	Number of capsule	Vector dimension	1D-CapsGAN			2D-CapsGAN		
			OA(%)	AA(%)	KAPPA(%)	OA(%)	AA(%)	KAPPA(%)
3	8	8	**74.38**	**83.56**	**68.25**	96.44	97.67	95.31
		16	67.75	79.21	60.01	97.21	97.22	96.30
5	8	8	68.46	76.42	61.52	96.47	97.10	95.34
		16	71.27	78.51	64.13	97.52	97.74	96.72
7	8	8	60.16	75.67	53.00	94.84	96.18	93.21
		16	64.07	72.86	55.92	97.49	97.40	96.67
3	16	8	61.14	77.19	53.41	96.58	97.40	95.50
		16	66.67	76.06	58.95	96.43	97.14	95.30
5	16	8	63.76	75.55	56.58	96.60	96.65	95.50
		16	67.19	79.51	60.10	96.5	97.57	95.42
7	16	8	62.01	74.19	54.28	96.23	96.88	95.04
		16	71.83	82.20	65.09	94.53	96.13	92.83
3	32	8	62.91	71.72	54.70	**97.53**	**97.66**	**96.73**
		16	67.82	77.23	60.68	95.74	97.02	94.40
5	32	8	67.09	74.60	59.76	96.88	97.37	95.88
		16	64.74	75.17	57.33	95.19	97.02	93.71
7	32	8	66.65	79.16	59.84	96.08	97.06	94.84
		16	56.90	73.50	49.58	96.57	96.95	95.46

TABLE V
CLASSIFICATION EFFECT OF D_SPE AND D_SPA UNDER DIFFERENT PARAMETERS FOR THE SALINAS DATA SET

Filter size	Number of capsule	Vector dimension	1D-CapsGAN			2D-CapsGAN		
			OA(%)	AA(%)	KAPPA(%)	OA(%)	AA(%)	KAPPA(%)
3	8	8	82.22	90.91	80.43	92.85	96.80	92.05
		16	74.61	84.38	72.34	92.56	97.03	91.75
5	8	8	83.95	91.98	82.28	92.85	97.18	92.07
		16	84.13	91.79	82.49	92.26	96.67	91.41
7	8	8	83.17	90.40	81.42	92.60	96.86	91.79
		16	83.25	91.34	81.53	92.39	96.83	91.56
3	16	8	82.45	90.19	80.63	92.53	96.98	91.71
		16	81.63	90.03	79.77	91.84	96.42	90.95
5	16	8	**84.58**	**91.57**	**82.90**	**93.31**	**97.10**	**92.57**
		16	80.82	90.50	80.00	92.17	96.76	91.32
7	16	8	80.89	88.17	78.88	92.34	96.52	91.55
		16	83.79	90.84	82.11	92.57	96.93	91.75
3	32	8	80.00	86.33	77.92	92.50	96.91	91.68
		16	83.77	90.42	82.04	91.87	96.74	90.99
5	32	8	77.17	87.19	75.09	91.83	96.62	90.94
		16	82.53	91.00	80.82	92.05	96.64	91.18
7	32	8	81.15	88.50	79.24	91.91	96.59	91.02
		16	82.10	88.71	80.2	91.98	96.48	91.10

these SS features provide more efficient information for the classification of HSI data. Unlike CNN mapping features to scalars, CapsNet learns more discriminative features and the relative position of pixels by mapping features into vectors, which can be used to guide classification, so CapsNet outperforms 2-D-CNN and 3-D-CNN. GAN can augment more samples and produce satisfactory results. Because CapsNet and GAN are based on 2-D-CNN, and 3-D-CNN can fully learn the spectral feature compared with the traditional 2-D-CNN, so 3-D-CNN-based SSRN outperforms CapsNet and GAN. Under the guidance of the best network parameters explored in Section IV-B, our proposed algorithms DcCapsGAN outperforms the other competitive algorithms on all three data sets, achieving better than 97% using 40 samples per class as training for the UP data set, achieving better than 93% using 20 training samples per class for the Salinas image, achieving better than 98% using 25 training samples per class for the KSC data set, and the proposed DCCapsGAN achieves the best classification results in the majority of classes. All these accuracy results verified the effectiveness of our proposed

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TABLE VI
CLASSIFICATION EFFECT OF D_SPE AND D_SPA UNDER DIFFERENT PARAMETERS FOR THE KSC DATA SET

Filter size	Number of capsule	Vector dimension	1D-CapsGAN			2D-CapsGAN		
			OA(%)	AA(%)	KAPPA(%)	OA(%)	AA(%)	KAPPA(%)
3	8	8	81.31	74.53	79.19	97.44	96.31	97.14
		16	81.62	72.70	79.45	97.26	96.33	96.94
5	8	8	82.60	75.83	80.56	96.93	95.44	96.57
		16	82.83	76.73	80.84	96.85	95.11	96.48
7	8	8	83.18	78.70	81.22	96.48	94.68	96.07
		16	**85.22**	**79.69**	**83.49**	97.36	96.14	97.05
3	16	8	81.42	73.35	79.23	97.61	96.15	97.32
		16	81.72	69.74	79.55	97.34	95.75	97.03
5	16	8	83.14	76.31	81.13	**97.63**	**96.52**	**97.35**
		16	83.59	74.68	81.65	97.34	96.41	97.03
7	16	8	84.24	79.63	82.41	97.18	95.93	96.83
		16	84.06	79.61	82.18	96.58	94.32	96.18
3	32	8	83.48	76.89	81.56	96.56	95.17	96.16
		16	83.36	78.49	81.41	96.83	95.13	96.46
5	32	8	84.12	76.38	82.24	95.87	95.61	95.39
		16	84.79	80.35	83.03	97.34	95.88	97.03
7	32	8	84.45	79.29	82.62	96.32	94.38	95.89
		16	83.75	78.78	81.85	97.22	95.45	96.89

TABLE VII
ABLATION STUDY ON THE PAVIA UNIVERSITY DATA SET

methods	OA(%)	Training time(s)
2D-CapsGAN	96.95	32.60
exclude OctConv	95.51	39.19
exclude multiConv	96.12	32.29
1D-CapsGAN	78.79	57.47
exclude multiConv	74.19	34.29

dual-channel capsule and GAN structure combination strategy. From Tables VIII–X, we can also observe that the running speeds of 2-D-CNN, 3-D-CNN, and GAN are much faster but present low accuracies. The proposed DcCapsGAN is mainly based on CapsNet for exploiting more efficient spatial and spectral relations by replacing the scalar feature values in traditional CNN with vectors; therefore, it brings a lot of parameters. However, it still presents higher accuracy and more efficiency than current the state-of-the-art algorithm, SSRN.

The classification maps of different algorithms for three data sets are illustrated in Figs. 11–13. We can observe from Figs. 11(c)–13(c) that there are many discrete misclassification points in the classification map of SVM due to only exploit spectral information. Considering that the spatial information is considered in 2-D-CNN, the classification map obtained a more accurate and smooth result than SVM. For jointly considering SS information in SVMCK and 3-D-CNN, which can alleviate the oversmooth of 2-D-CNN, compared with other comparison algorithms, SSRN presents smoother result because of considering spatial features and continuous spectral features, but it also shows serious oversmooth, such as the area marked by the yellow box in Fig. 11(g). Moreover, CapsNet represents the features as vectors and

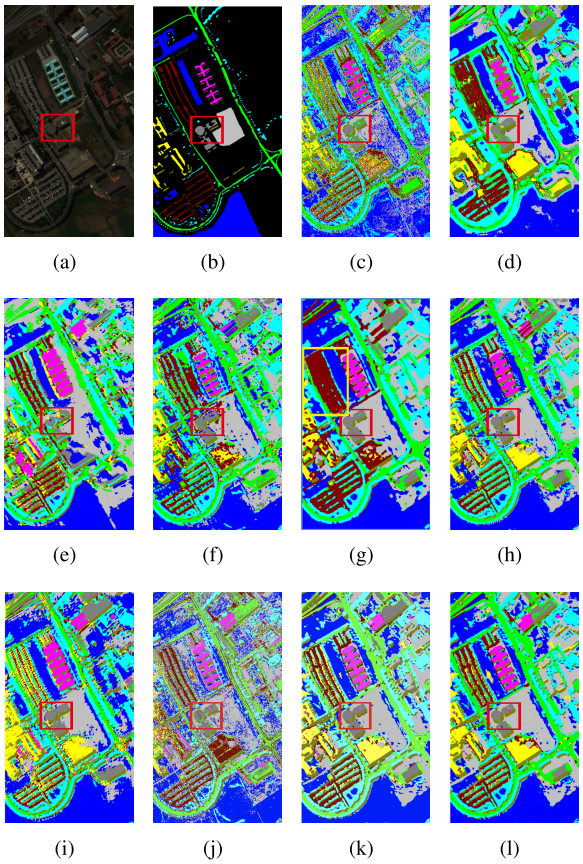


Fig. 11. Classification maps for the UP data set. (a) Pseudocolor composite image. (b) Ground truth. (c) SVM. (d) SVMCK. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) CapsNet. (i) GAN. (j) 1-D-CapsGAN. (k) 2-D-CapsGAN. (l) DcCapsGAN.

mines the relative position information of samples to produce a more realistic classification map, while the classification map of GAN is smoother. Compared with the

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TABLE VIII
CLASSIFICATION RESULTS FOR THE UP DATA SET

class	SVM	SVMCK	2D-CNN	3D-CNN	SSRN	CapsNet	GAN	1D-CapsGAN	2D-CapsGAN	DcCapsGAN
1	72.04±1.48	86.04±3.07	85.36±3.99	91.25±0.66	**99.57±0.22**	94.88±0.93	79.88±1.66	70.26±1.87	94.47±1.82	97.80±2.37
2	76.96±5.21	90.97±3.81	79.59±4.69	76.07±5.25	98.40±0.18	93.49±5.44	97.00±1.83	76.00±3.81	97.84±1.44	**99.74±0.13**
3	74.58±5.35	93.89±1.86	67.25±3.76	82.89±7.12	**95.50±5.18**	87.19±2.86	91.19±5.92	50.89±0.11	92.27±0.13	93.49±3.76
4	92.24±3.39	96.11±0.95	85.13±5.15	96.08±1.03	98.98±0.15	97.88±0.58	98.41±1.38	91.67±0.47	98.49±0.02	**99.42±0.55**
5	98.97±0.47	99.72±0.11	97.98±2.74	99.10±0.62	99.82±0.25	99.97±0.05	99.92±0.08	99.92±0.18	**100±0.00**	**100±0.00**
6	74.96±4.79	93.09±1.83	64.14±10.51	89.59±0.40	94.17±6.14	97.06±1.47	97.86±1.83	85.90±1.48	**98.97±0.17**	98.12±1.12
7	90.49±3.11	97.93±0.51	75.35±3.05	99.38±0.13	91.81±11.96	95.99±4.69	96.41±2.10	92.24±0.14	**99.76±0.00**	97.98±2.31
8	76.17±7.54	90.74±4.56	88.27±3.90	88.91±3.48	86.11±8.72	91.88±0.66	79.44±6.64	86.27±0.51	92.68±1.76	**96.83±0.42**
9	98.25±1.63	96.03±1.50	96.43±0.66	99.63±0.46	99.23±0.97	99.78±0.11	99.78±0.11	99.66±0.15	**99.88±0.00**	99.85±0.13
OA(%)	78.41±1.92	91.53±2.15	80.01±3.30	83.77±0.46	97.02±0.44	94.41±2.44	92.92±2.34	78.79±1.12	96.95±0.21	**97.98±0.14**
AA(%)	83.85±0.95	93.84±0.89	82.17±2.57	91.44±0.66	96.15±1.22	95.35±1.13	93.36±1.40	83.65±1.62	97.15±0.23	**98.01±0.22**
KAPPA(%)	72.34±2.23	88.94±2.74	74.33±4.04	80.75±2.72	96.06±0.58	92.67±3.12	90.68±3.04	73.06±1.72	95.97±0.27	**97.32±0.18**
Training time(s)	-	-	4.54	3.86	250.56	11.73	9.07	57.47	32.60	65.30

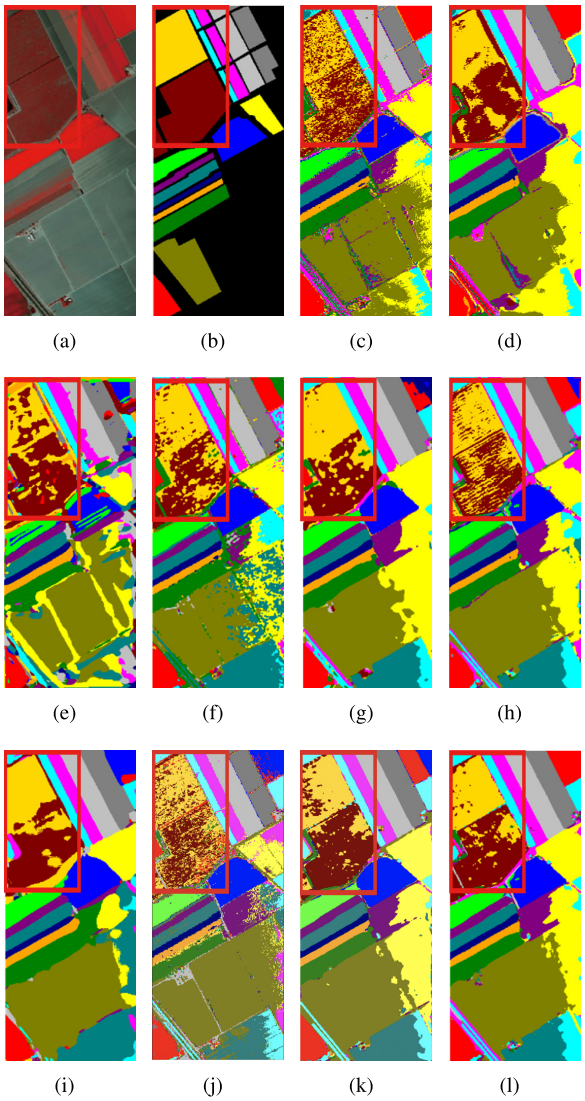


Fig. 12. Classification maps for the Salinas data set. (a) Pseudocolor composite image. (b) Ground truth. (c) SVM. (d) SVMCK. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) CapsNet. (i) GAN. (j) 1-D-CapsGAN. (k) 2-D-CapsGAN. (l) DcCapsGAN.

abovementioned comparison algorithm, the proposed DcCaps-GAN combines the advantages of CapsNet and GAN and appears to provide more consistent classification results.

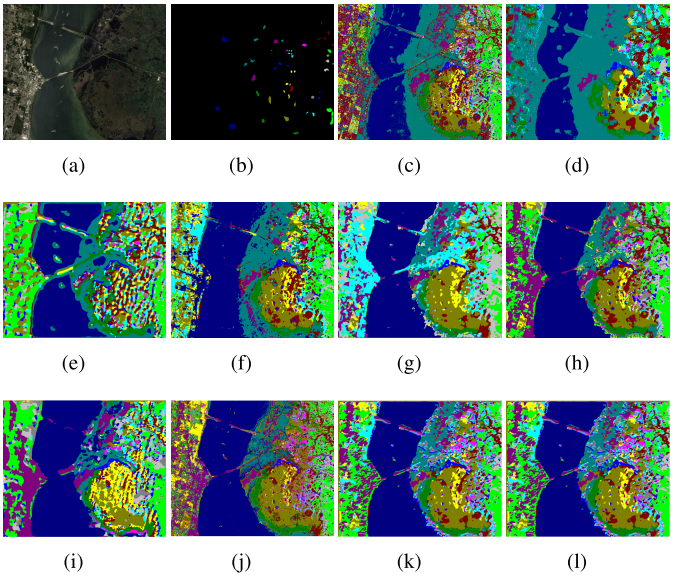


Fig. 13. Classification maps for the KSC data set. (a) Pseudocolor composite image. (b) Ground truth. (c) SVM. (d) SVMCK. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) CapsNet. (i) GAN. (j) 1-D-CapsGAN. (k) 2-D-CapsGAN. (l) DcCapsGAN.

In particular, in the small area marked by the red box rectangle region in Fig. 11 and the yellow rectangle region in Fig. 12, we can observe that our proposed structure achieves the most satisfying results in both edge detection and classification performance.

D. Effect of Different Number of Training Samples

As shown in Fig. 14, the OA results under different training sample rates are illustrated. Especially, in each of the three data sets, ten, 20, 30, and 40 samples were randomly selected for training. Compared with SVM, SVMCK, 2-D-CNN, 3-D-CNN, SSRN, CapsNet, and GAN, the proposed DcCaps-GAN shows the best performance. The DL-based method presents better performance than SVM, and the effect of the spatial-based 2-D-CNN is lower than that of the SS-based 3-D-CNN, which can prove that DL and SS information are critical to classification. With the increase in the number of training samples, it is apparent that the performance of all

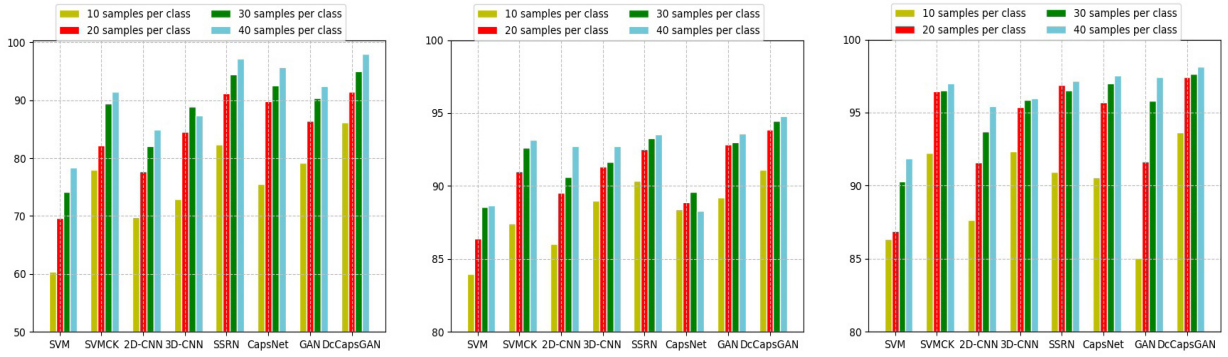
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TABLE IX
CLASSIFICATION RESULTS FOR THE SALINAS DATA SET

class	SVM	SVMCK	2D-CNN	3D-CNN	SSRN	CapsNet	GAN	1D-CapsGAN	2D-CapsGAN	DcCapsGAN
1	97.93±0.78	98.17±1.40	76.97±4.45	**100±0.00**	**100±0.00**	99.59±0.53	99.50±0.87	98.89±1.42	99.94±0.03	99.40±0.96
2	98.50±0.21	98.81±0.98	71.34±7.01	99.60±0.59	99.46±0.81	99.33±0.92	99.24±0.36	96.77±3.98	**99.89±0.04**	99.76±0.03
3	92.07±3.58	92.63±6.27	83.86±7.97	99.66±0.2	98.20±1.67	99.66±0.30	99.85±0.14	92.51±5.31	99.94±0.01	**99.98±0.03**
4	99.29±0.37	99.36±0.11	99.25±0.63	94.42±1.62	97.29±2.58	99.39±0.28	99.85±0.25	99.63±0.00	99.78±0.05	**100±0.00**
5	96.01±2.68	96.55±4.30	83.70±7.35	95.55±2.25	**99.32±0.87**	98.61±1.23	98.66±0.74	92.45±6.14	95.65±0.13	96.72±1.53
6	99.59±0.20	99.90±0.07	92.67±3.04	99.69±0.54	99.93±0.02	99.67±0.40	99.90±0.17	99.70±0.05	99.82±0.03	**100±0.00**
7	99.26±0.04	98.96±0.91	78.90±7.47	99.71±0.08	99.90±0,07	99.46±0.28	98.59±2.22	99.73±0.09	99.83±0.19	**100±0.00**
8	68.82±3.17	73.57±2.98	69.15±8.73	70.11±5.17	**89.24±8.73**	70.58±8.78	76.77±5.06	66.79±6.92	80.01±2.67	77.10±4.55
9	96.28±0.23	95.73±1.12	83.47±6.19	99.96±0.02	99.60±0.25	99.94±0.04	99.99±0.01	98.22±1.98	99.90±0.02	**99.99±0.01**
10	87.54±5.97	95.00±0.98	90.57±2.46	95.06±2.06	95.86±1.44	95.38±1.52	**98.14±0.59**	82.25±4.42	97.39±0.13	96.65±0.77
11	95.57±3.37	99.68±0.21	97.71±0.94	95.61±2.01	94.53±2.13	98.12±1.87	99.78±0.24	93.17±1.41	99.76±0.06	**99.93±0.11**
12	99.65±0.51	99.30±0.08	95.24±2.72	99.96±0.03	99.91±0.15	99.58±0.50	99.16±1.40	97.35±3.67	99.89±0.03	**99.97±0.06**
13	97.87±0.11	99.44±0.11	97.70±2.12	99.74±0.23	99.48±0.53	99.74±0.17	98.47±1.00	97.60±0.39	99.88±0.05	**99.96±0.06**
14	92.82±4.01	97.61±2.50	94.35±5.84	94.54±3.50	96.78±2.39	97.90±1.59	99.34±0.76	90.52±1.14	98.66±0.13	**99.97±0.06**
15	65.36±7.38	88.58±5.72	69.41±12.87	82.62±4.42	75.19±8.85	76.81±6.55	87.51±5.55	53.41±2.56	84.57±3.47	**93.57±2.17**
16	93.71±4.11	88.22±2.24	91.10±5.89	98.77±0.44	**99.98±0.03**	97.87±0.84	99.09±1.40	95.71±0.91	99.80±0.03	99.31±0.67
OA(%)	86.39±1.18	90.99±0.56	80.13±0.19	90.44±0.67	93.05±0.83	90.05±1.85	93.01±0.63	83.95±0.32	93.27±0.12	**93.87±0.82**
AA(%)	92.52±0.86	95.12±0.22	85.96±0.39	95.31±0.28	96.58±0.62	95.73±0.59	97.16±0.15	90.92±0.41	97.17±0.01	**97.54±0.33**
KAPPA(%)	84.87±1.31	90.01±0.62	78.04±0.16	89.39±0.73	92.27±0.92	88.96±2.04	92.25±0.71	82.20±0.42	92.53±0.12	**93.20±0.90**
Training time(s)	-	-	5.06	4.05	286.28	19.21	8.87	76.54	70.13	167.54

TABLE X
CLASSIFICATION RESULTS FOR THE KSC DATA SET

class	SVM	SVMCK	2D-CNN	3D-CNN	SSRN	CapsNet	GAN	1D-CapsGAN	2D-CapsGAN	DcCapsGAN
1	89.53±3.91	98.09±0.11	84.28±1.10	95.38±3.12	98.74±2.18	99.50±0.16	96.83±1.02	89.26±2.11	99.32±0.57	**99.54±0.08**
2	85.16±6.80	98.16±0.91	92.51±6.36	91.74±4.20	95.45±5.00	97.25±3.31	95.87±5.58	86.23±1.94	99.77±0.32	**100±0.00**
3	91.63±2.38	99.35±0.64	96.83±1.75	93.65±1.32	98.16±1.12	95.53±2.88	95.24±3.38	67.53±7.95	98.26±1.83	**98.99±0.25**
4	74.30±2.42	86.12±3.30	92.51±2.45	79.30±9.81	**94.81±6.32**	63.29±6.91	87.37±1.99	53.08±9.65	71.14±2.80	81.94±1.16
5	75.00±5.09	88.23±10.29	99.75±0.43	90.69±2.97	79.56±3.83	90.44±3.37	**99.75±0.43**	69.48±2.31	94.48±1.55	90.69±0.43
6	59.31±6.73	93.13±2.94	92.98±2.42	94.94±0.75	88.51±1.9	91.83±1.02	98.20±1.72	66.66±4.85	97.05±2.07	**98.37±0.28**
7	95.00±4.33	**100±0.00**	**100±0.00**	97.08±3.15	78.05±1.9	**100±0.00**	**100±0.00**	78.75±3.53	**100±0.00**	**100±0.00**
8	88.66±3.44	99.87±0.12	98.61±2.00	96.71±1.00	**100±0.00**	98.28±1.70	93.68±4.44	79.18±6.44	99.50±0.69	**100±0.00**
9	95.62±3.03	99.69±0.30	87.75±4.35	94.28±6.41	99.44±0.95	99.60±0.00	92.46±5.16	98.48±1.28	99.59±0.00	**99.73±0.23**
10	92.61±0.91	98.94±1.05	99.47±0.27	97.45±3.97	99.74±0.00	99.56±0.16	93.93±3.55	95.91±0.18	99.60±0.18	**100±0.00**
11	92.97±1.68	93.90±5.32	92.72±4.80	93.23±5.86	**100±0.00**	98.98±0.67	99.92±0.14	90.22±0.53	97.20±0.00	97.29±0.82
12	81.86±2.33	99.79±0.20	94.56±2.19	**99.16±1.27**	99.14±1.48	97.56±1.42	97.49±0.73	77.09±0.44	97.80±0.73	97.98±0.12
13	97.59±0.33	99.39±0.60	99.93±0.13	99.85±0.25	99.78±0.00	99.85±0.25	**100±0.00**	99.94±0.07	99.77±0.00	**100±0.00**
OA(%)	88.95±0.88	97.59±0.68	93.97±0.81	95.45±0.79	96.77±1.65	96.70±0.25	96.43±0.58	86.22±0.23	97.59±0.13	**98.20±0.06**
AA(%)	86.10±0.84	96.51±1.29	94.76±0.76	94.11±0.73	94.76±1.28	94.74±0.31	96.21±0.84	80.91±1.82	96.42±0.24	**97.23±0.07**
KAPPA(%)	89.53±0.01	97.31±0.76	93.28±0.90	94.92±0.88	96.39±1.84	96.31±0.28	96.02±0.65	84.60±0.27	97.31±0.14	**97.99±0.07**
Training time(s)	-	-	4.79	4.00	278.70	16.16	8.97	38.63	74.59	126.17



(a) (b) (c)

Fig. 14. OA results with different number of training samples. (a) On UP data set. (b) On Salinas data set. (c) On KSC data set.

the competitive algorithms and the proposed DcCapsGAN improved as the size of the training sets. DcCapsGAN consistently yields higher OA than any other algorithms and presents a more stable and satisfying performance even under limited available training sample situations.

V. CONCLUSION

In this article, we proposed a dual-channel SS fusion capsule generative adversarial network (DcCapsGAN) with simultaneously considering the augmentation of training samples and encoding a more feature relationship with capsule structure.

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The proposed DcCapsGAN can effectively exploit and fusion the SS information by considering spectral and spatial contexture relations, where CapsNet structure is constructed in the discriminator part of the GAN to maintain the SS details of the samples and improve the stability of the proposed model. Meanwhile, octave convolution and multiscale convolution are migrated to the proposed model to construct and reduce the parameters of the CapsNet with simultaneously extracting multiscale information, in terms of effectively improving the classification efficiency and accuracy of the proposed framework.

Although the proposed DcCapsGAN presents very encouraging results in classification accuracy, the computational cost increases by involving capsule structure. As a consequence, some algorithms for further reducing the computation cost by increasing the fidelity of augmented samples will be taken into consideration in our future work.

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